

CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

Lab Solutions – Set 7

Question 13

Write a Pandas program to detect missing values of a given DataFrame and display True or False.

Python Code:

```
import pandas as pd
import numpy as np

# Create sample DataFrame with missing values
data = {
    'ORD_NO': [70001, np.nan, 70002, 70004, np.nan, 70005],
    'PURCH_AMT': [150.5, 270.65, 65.26, 110.5, 948.5, 2400.6],
    'ORD_DATE': ['2012-10-05', '2012-09-10', np.nan, '2012-08-17', '2012-09-10',
                 '2012-07-27'],
    'CUSTOMER_ID': [3002, 3001, 3001, 3006, 3002, 3001]
}

df = pd.DataFrame(data)

# Detect missing values (True for missing, False otherwise)
missing_val_df = df.isna()

print("Original DataFrame:")
print(df)
print("\nMissing values detected (True/False):")
print(missing_val_df)
```

Output:

```
Original DataFrame:
   ORD_NO  PURCH_AMT  ORD_DATE  CUSTOMER_ID
0  70001.0    150.50  2012-10-05          3002
1     NaN    270.65  2012-09-10          3001
2  70002.0     65.26           NaN          3001
3  70004.0    110.50  2012-08-17          3006
4     NaN    948.50  2012-09-10          3002
5  70005.0   2400.60  2012-07-27          3001

Missing values detected (True/False):
   ORD_NO  PURCH_AMT  ORD_DATE  CUSTOMER_ID
0  False     False     False     False
1   True     False     False     False
2  False     False     True      False
3  False     False     False     False
4   True     False     False     False
5  False     False     False     False
```

Question 14

Write a Pandas program to find and replace the missing values in a given DataFrame that do not have any valuable information.

Python Code:

```
import pandas as pd
import numpy as np

# Create sample DataFrame with missing values
data = {
    'ORD_NO': [70001, np.nan, 70002, 70004, np.nan, 70005],
    'PURCH_AMT': [150.5, 270.65, 65.26, 110.5, 948.5, 2400.6],
    'ORD_DATE': ['2012-10-05', '2012-09-10', np.nan, '2012-08-17', '2012-09-10',
                 '2012-07-27'],
    'CUSTOMER_ID': [3002, 3001, 3001, 3006, 3002, 3001]
}

df = pd.DataFrame(data)

print("Original DataFrame with Missing Values:")
print(df)

# Find and replace missing values
df_replaced = df.fillna({'ORD_NO': 0, 'ORD_DATE': 'Missing'})

print("\nDataFrame after Replacing Missing Values:")
print(df_replaced)
```

Output:

```
Original DataFrame with Missing Values:
   ORD_NO  PURCH_AMT  ORD_DATE  CUSTOMER_ID
0  70001.0    150.50  2012-10-05          3002
1      NaN    270.65  2012-09-10          3001
2  70002.0     65.26          NaN          3001
3  70004.0    110.50  2012-08-17          3006
4      NaN    948.50  2012-09-10          3002
5  70005.0   2400.60  2012-07-27          3001

DataFrame after Replacing Missing Values:
   ORD_NO  PURCH_AMT  ORD_DATE  CUSTOMER_ID
0  70001.0    150.50  2012-10-05          3002
1     0.0    270.65  2012-09-10          3001
2  70002.0     65.26    Missing          3001
3  70004.0    110.50  2012-08-17          3006
4     0.0    948.50  2012-09-10          3002
5  70005.0   2400.60  2012-07-27          3001
```